Residuals Matter, Noise Still Wins: Insights into Plant-Soil Segmentation

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***Abstract*—We compared classical baselines and deep learning models for plant-soil segmentation on the EWS dataset. The methods include ExG, Logistic Regression, Random Forest, U-Net, ResU-Net, ASPP-ResU-Net, and Attention-Gated ASPP-ResU-Net. All deep models surpassed the Plant IoU benchmark of 0.775 reported in the original paper, with the best model reaching 0.8327. The largest clean-set improvement appeared when moving from U-Net to ResU-Net, while ASPP and attention gates provided smaller additional gains. However, the robustness results revealed a more nuanced picture. Under strong Gaussian noise, all residual-based models collapsed near zero Plant IoU, while the plain U-Net retained a Plant IoU of 0.27 at the highest severity—a complete ranking reversal relative to clean-set performance. This suggests a meaningful accuracy-robustness trade-off: the residual-based designs that improve clean-set accuracy in our setting also appear substantially more vulnerable to severe Gaussian noise.**

***Index Terms*—Plant-soil segmentation, semantic segmentation, deep learning, robustness, U-Net, residual learning.**

I. Introduction

Plant-soil segmentation is an important task in precision agriculture because it supports later analysis such as crop monitoring, field measurement, and automated decision-making. In real field scenes, however, this task is not as simple as it first appears. Compared with controlled laboratory images, field images are much harder to handle because lighting changes constantly, soil appearance is often uneven, and environmental conditions differ across time and location. As a result, even binary segmentation can become difficult in practice.

In this project, we use the EWS Dataset released through the ETH Research Collection [1]. The dataset contains RGB field images together with pixel-level masks for plant-soil segmentation. This provides a useful setting for comparing different methods under the same conditions. More importantly, it allows us to examine how these methods behave not only when the images are clean, but also when the visual conditions become less ideal.

Based on this setting, the main question of the project is straightforward: which kinds of methods are most effective for plant-soil segmentation in realistic field conditions? To answer this question, we compare both classical baselines and deep learning models within a unified experimental pipeline. We evaluate them using the same preprocessing and metrics, and we consider both clean-set performance and robustness under corruption. In this way, the project is not only about identifying the best-performing model, but also about understanding where the improvement actually comes from.

II. Literature Review

Earlier segmentation methods in agricultural vision often relied on handcrafted rules. A common example is the use of vegetation indices such as Excess Green (ExG), sometimes combined with thresholding or simple classifiers. These methods are still useful because they are easy to implement, interpretable, and computationally cheap. However, they usually depend heavily on stable colour contrast between vegetation and background. Once illumination changes or the soil becomes visually complex, their performance can drop quickly. For this reason, they are helpful as baselines, but they are usually not strong enough on their own for more difficult field conditions.

With the development of deep learning, segmentation methods gradually shifted toward representation-based models. Encoder-decoder architectures, especially U-Net and its variants, became widely used because they can learn pixel-level segmentation directly from data while still preserving spatial detail through skip connections. This is important in agricultural images, where plant regions may be small, irregular, or visually close to the background. In these cases, the model needs more than colour information alone; it also needs to use local texture, boundary cues, and surrounding context.

Later studies extended this general framework in several ways. Residual connections were introduced to improve optimization and feature reuse, especially in deeper networks. Multi-scale modules such as ASPP were designed to capture information at different receptive-field sizes, while attention mechanisms were used to guide the model toward more relevant regions and suppress background interference. Although these ideas are different in form, they are all trying to solve a similar problem: how to make segmentation more accurate and more reliable in visually complicated scenes.

Recent work in plant phenotyping also shows that model performance depends strongly on real-world variability. Deep learning methods have already shown strong results for outdoor plant segmentation under diverse field conditions [2]. More recent studies have further explored how to improve crop segmentation through stronger supervision and additional geometric information, such as depth-informed pseudo-labelling and large-scale depth-labelled data [3], [4]. These studies suggest that high clean-set accuracy is important, but robustness under realistic variation is just as important.

Following this line of work, our project adopts a progression from simple baselines to more expressive deep models. This design is useful because it allows us to compare not only which model performs best, but also why the performance changes. In other words, we are interested in whether the gain mainly

comes from handcrafted colour knowledge, stronger nonlinear classification, or better learned spatial representations.

III. Methods

We treat the task as binary semantic segmentation, where each pixel in a field image is labelled as plant or soil. This pixel-level setting is appropriate for agricultural applications that need accurate estimates of plant coverage and boundaries [5]. All images and masks are resized to 352×352 so that every method is tested under the same conditions.

Our goal is not only to rank a set of models, but to understand why some of them work better than others. For that reason, the experimental design moves from simple and interpretable baselines to models with increasingly strong representational power. This progression helps reveal whether the main improvement comes from handcrafted colour knowledge, stronger nonlinear classification, or learned spatial features that make better use of image context.

A. Classical Methods

We begin with classical methods because they represent the simplest way to approach the problem. If plant and soil can already be separated reasonably well using handcrafted colour information, then the value of more complex models needs to be demonstrated rather than assumed. For this reason, the classical baselines are not included merely as weak references, but as an important first step in understanding where the real difficulty of the task lies.

The most direct classical baseline is Excess Green (ExG), computed as $2G - R - B$. ExG is widely used as a simple vegetation prior because it emphasizes green-dominant regions in RGB imagery [6]. We include it not only because it is easy to implement, but because it asks an important question at the start of the study: how much of the task can already be solved by an explicit color rule? In our experiments, ExG is used both as a standalone zero-training baseline and as an additional handcrafted feature in later ablations, which also lets us test whether such a prior remains useful once more expressive models are introduced.

After ExG, we introduce Logistic Regression as the first learned pixel-wise baseline. Unlike ExG, which relies on a fixed color rule, Logistic Regression learns a linear decision boundary from handcrafted features [7]. Each pixel is treated as a sample, and we test three feature sets: RGB, RGB+HSV, and RGB+HSV+ExG. HSV is included because it separates chromatic content from brightness and may help when illumination is less stable [8]. The model is trained with `max_iter=1000`, balanced class sampling, and a fixed random seed of 42. This gives us a simple but informative reference for what can be achieved through feature engineering and linear classification alone.

We then introduce Random Forest as a stronger nonlinear baseline. Compared with Logistic Regression, Random Forest can capture more complex interactions among handcrafted features because it combines an ensemble of decision trees [9]. To keep the comparison consistent, we use the same three feature settings—RGB, RGB+HSV, and RGB+HSV+ExG—and train the model with `n_estimators=100`, `max_depth=15`, `samples_per_class=2000`, and `random_state=42`. This allows us to see how much is gained by replacing a linear classifier with a more flexible nonlinear one, while still staying within the handcrafted-feature setting.

Together, the classical baselines define the point at which handcrafted colour information begins to reach its limit. They remain attractive because they are simple and interpretable, but their reliance on manually designed features also makes them sensitive to changing field conditions. This naturally leads to the next step of the study: testing whether deep models can move beyond that limitation by learning stronger spatial representations directly from the images.

B. Deep Learning Segmentation Models

Once the comparison moves past the classical baselines, the central question changes slightly. The issue is no longer whether a handcrafted feature can separate plant from soil, but whether learned spatial representations can do so more reliably under realistic field variation. For this reason, the deep-learning part of the study focuses on encoder-decoder architectures, which are designed to combine local detail with broader contextual information.

The natural starting point for the deep-model comparison is **U-Net**. It provides a clear baseline because it already captures the essential encoder-decoder idea: semantic abstraction in the contracting path, spatial recovery in the expanding path, and skip connections that preserve fine detail across the network [10]. In this study, U-Net is not treated as a weak baseline, but as the model that defines the standard against which all later improvements need to justify themselves.

The first major refinement to the U-Net baseline is **ResU-Net**, which introduces residual learning into the encoder-decoder structure. This is a particularly meaningful step in the study because residual connections are not mainly about making the architecture more complicated; they are about making learning more stable and feature propagation more effective [11]. In our implementation, the residual blocks are paired with ReLU activations and Group Normalization to suit the small-batch training setting [14]. Conceptually, ResU-Net tests whether the main bottleneck of the baseline is poor optimization rather than insufficient architectural sophistication. This later becomes one of the clearest patterns in the results.

The next model, **ASPP-ResU-Net**, builds on the residual backbone by introducing **Atrous Spatial Pyramid Pooling (ASPP)**. ASPP is designed to aggregate information across multiple receptive fields, allowing the model to combine local structure with larger-scale context [12]. This is a meaningful extension in field imagery, where plant-soil separation may depend not only on local appearance but also on surrounding texture and regional context. In this sense, ASPP-ResU-Net asks whether multi-scale contextual modelling can further improve a backbone that is already strong in optimization and feature reuse.

The final model in the deep-learning comparison is **Attention-Gated ASPP-ResU-Net**, which adds attention-based skip refinement to the ASPP-ResU-Net architecture.

TABLE I: Summary of Implemented Methods

Method	Type	Key design
ExG	Classical	Colour index, zero training
Logistic Reg.	Classical ML	Linear, RGB/HSV/ExG
Random Forest	Classical ML	Ensemble, nonlinear
U-Net	Deep	Encoder-decoder, skip connections
ResU-Net	Deep	+ Residual blocks, GroupNorm
ASPP-ResU-Net	Deep	+ Multi-scale atrous pooling
AG-ASPP-ResU-Net	Deep	+ Attention-gated skip fusion

Attention mechanisms help the model emphasize informative regions while suppressing less relevant background responses [13]. Here, attention gating is used to make skip fusion more selective, with the goal of improving foreground-background separation in cluttered field scenes. This model therefore represents the most refined architecture in the study: not simply deeper or larger, but more selective in how information is propagated through the network.

To conclude, the four deep architectures are not simply a list of alternatives. They represent a structured sequence of hypotheses: U-Net tests the baseline encoder-decoder paradigm; ResU-Net tests whether better optimization and feature reuse improve the task; ASPP-ResU-Net tests whether multi-scale context matters; and Attention-Gated ASPP-ResU-Net tests whether more selective feature fusion can refine the strongest model further. Table I summarises all implemented methods.

C. Training Configuration and Implementation Choices

All deep models are trained under the same protocol so that differences in performance can be attributed to architecture rather than inconsistent optimization. Input images are normalized to $[0, 1]$, and the segmentation masks are converted into binary form with plant pixels treated as the positive class.

The loss function combines **binary cross-entropy with logits** and **Dice loss** in equal proportion. This choice reflects the fact that the task requires both stable pixel-wise learning and strong overlap quality between prediction and ground truth. Binary cross-entropy supports optimization at the pixel level, while Dice loss directly emphasizes region-level agreement, which is particularly useful in segmentation settings with foreground-background imbalance.

Optimization is carried out using **AdamW** with an initial learning rate of 1×10^{-4}. Learning rate reduction on validation plateau and early stopping are both applied so that the comparison reflects the best achievable model state rather than arbitrary final-epoch behaviour. In the main architecture comparison, the batch size is set to 3, the maximum number of epochs is 100, and the inference threshold is 0.6. A fixed random seed of 42 is used throughout the experiments. During training, checkpoints are saved and the best-performing checkpoint on validation loss is used for final evaluation.

The framework also supports feature-augmented inputs, including ExG and HSV-derived channels. These are introduced only in ablation experiments, allowing the study to test whether explicit agricultural priors still provide useful information once the model is already able to learn hierarchical representations directly from RGB input.

D. Evaluation Design and Robustness Protocol

Our evaluation is designed to do more than identify the highest clean-set score. All methods are first tested on the standard train/validation/test split, using the same preprocessing pipeline and the same resized image resolution. Performance is measured with class-wise IoU, precision, recall, F1-score, overall accuracy, confusion matrices, and inference time.

A second part of the evaluation focuses on robustness. Since field imagery is rarely collected under ideal conditions, the models are further tested under several controlled corruption types: **Gaussian noise**, **Gaussian blur**, **brightness shift**, **affine rotation**, and **JPEG compression**. Each corruption is applied at multiple severity levels so that degradation trends can be observed rather than inferred from a single perturbed setting.

This robustness-oriented design is central to the study. It allows the comparison to distinguish between models that perform well only under clean benchmark conditions and models that remain reliable when image quality changes. In this way, the evaluation framework supports a more practical interpretation of segmentation performance, one that is closer to real deployment conditions in agricultural environments.

IV. Experimental Results

A. Evaluation Protocol

All methods were evaluated under the same train/validation/test split and preprocessing pipeline described in Section III. We report class-wise IoU and F1-score, pixel accuracy, confusion matrices, and inference time. To examine reliability beyond the clean setting, robustness was further evaluated under controlled corruptions, including Gaussian noise, Gaussian blur, brightness shift, affine rotation, and JPEG compression, each tested at multiple severity levels. Gaussian noise was evaluated at five severity levels, with the highest level corresponding to variance 0.2.

B. Classical Baselines and Feature Effects

The classical results already show a clear pattern. Plant-soil separation can be achieved to a reasonable extent using handcrafted colour information, but the quality remains limited once the visual conditions become less ideal. Both Logistic Regression and Random Forest improve when moving beyond RGB-only features, and the gain from adding HSV is especially clear. However, the extra ExG channel brings only a small improvement, particularly for Logistic Regression. This suggests that explicit feature engineering is useful, but only up to a point.

These results confirm that handcrafted features remain useful when the model itself is relatively simple. At the same

TABLE II: Test-set performance of classical methods under different feature configurations.

Method	Feature set	Accuracy	Plant IoU	Plant F1	Time (s
Logistic Reg.	RGB	0.8689	0.7016	0.8247	0.0477
Logistic Reg.	RGB+HSV	0.8774	0.7130	0.8325	0.0982
Logistic Reg.	RGB+HSV+ExG	0.8774	0.7130	0.8325	0.1039
Random Forest	RGB	0.8575	0.6665	0.7999	1.1872
Random Forest	RGB+HSV	0.8597	0.6786	0.8085	1.1444
Random Forest	RGB+HSV+ExG	0.8676	0.6902	0.8167	1.1355

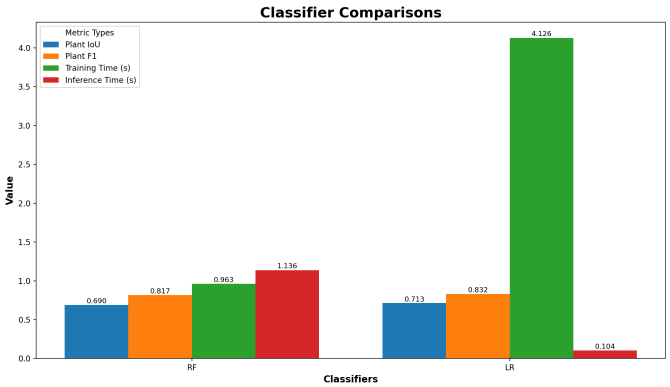


Fig. 1: Classical-method comparison. Plant IoU and F1 for Random Forest (RF) and Logistic Regression (LR) on the test set, together with training and inference time.

time, they also show the ceiling of that approach. Even the best classical configuration still falls well short of the deep models, which suggests that the main challenge in this task is not only learning better colour thresholds, but also learning spatial structure more effectively. ExG is therefore best understood as a strong zero-training baseline rather than a competitive alternative to the deep models.

C. Deep Model Comparison on Clean Splits

The deep model results tell a more structured story than a simple ranking of performance. All four encoder-decoder architectures clearly outperform the classical baselines, and all of them surpass the best Plant IoU of 0.775 reported in the original paper [2], with our strongest model reaching 0.8327—an improvement of 5.77 percentage points on this metric. The most important change within the deep-model progression, however, happens early. Moving from U-Net to ResU-Net produces the largest improvement, while ASPP and attention gating provide smaller but still consistent refinements. The best clean-set performance is achieved by Attention-Gated ASPP-ResU-Net, although the gap between the top three deep models is relatively small compared with the jump from U-Net to ResU-Net.

A closer comparison shows that the largest single gain came from moving from U-Net to ResU-Net, with Plant IoU increasing by about 1.29 percentage points. By contrast, ASPP and attention gating produced gains of 0.03 and 0.06 percentage points respectively. This suggests that residual learning contributed more to final performance than later architectural

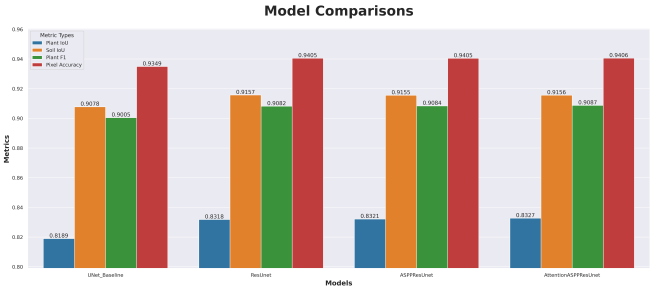


Fig. 2: Deep-model comparison on the test set. Plant IoU, Soil IoU, Plant F1 and pixel accuracy for U-Net, ResU-Net, ASPP-ResU-Net and Attention-Gated ASPP-ResU-Net. The largest gain occurs in the U-Net → ResU-Net step; subsequent ASPP and attention refinements add only marginal improvements.

refinements. The improvement from ASPP was especially limited, which is consistent with the nature of this task: in binary plant-soil segmentation, the scale variation is narrower than in more complex scene-segmentation problems, so the value of multi-scale receptive fields may be less pronounced.

Attention gating did not change the ranking dramatically, but it appears to have improved the balance between precision and recall on the plant class, partly reducing the false-negative tendency seen in earlier models. The validation scores are also consistently slightly higher than the test scores. For the best model, the validation-test gap in Plant IoU is about 1.2 percentage points. This gap is attributable to a year-based distribution shift: the training and validation sets contain images from 2018–2020, while the test set contains images from 2017, where illumination and soil conditions differ. This is consistent with the year variability analysis in the original paper [2], and reflects a stable domain shift rather than unstable training.

D. Training Behaviour and Confusion-Matrix Evidence

The training curves show that all deep models converge in a stable way. In later epochs, training loss continues to decrease while validation loss tends to flatten or fluctuate slightly, which is consistent with mild overfitting and supports the use of early stopping and best-checkpoint selection. In our experiments, the best checkpoint often appeared around epoch 40, after which training loss kept decreasing but validation performance no longer improved. This pattern is typical of small-dataset settings, where overfitting becomes visible before the nominal end of training. Figure 3 shows this pattern for the Attention-Gated ASPP-ResU-Net.

The confusion matrices tell a similar story from a different angle. The strongest models separate the two classes reliably, but the remaining mistakes are concentrated more on plant pixels than on soil pixels (Fig. 4). This suggests that the main difficulty is not general class confusion, but more specific ambiguity around plant boundaries and weak foreground structure.

E. Ablation and Negative Results

The ablation experiments are useful because they show that not every plausible modification leads to improvement.

TABLE III: Deep-model performance across train, validation, and test splits.

Model	Train Plant IoU	Val Plant IoU	Test Plant IoU	Test Plant F1	Test Soil IoU	Test Accuracy	Inference (s)
U-Net (Baseline)	0.7659	0.8317	0.8189	0.9005	0.9078	0.9349	1.0233
ResU-Net	0.8362	0.8424	0.8318	0.9082	**0.9157**	0.9405	1.1439
ASPP-ResU-Net	0.8390	0.8397	0.8321	0.9084	0.9155	0.9405	1.1977
AG-ASPP-ResU-Net	**0.8501**	**0.8447**	**0.8327**	**0.9087**	0.9156	**0.9406**	1.3510
Original paper best [2]	—	—	0.7750	0.8630	0.8990	0.9450	—

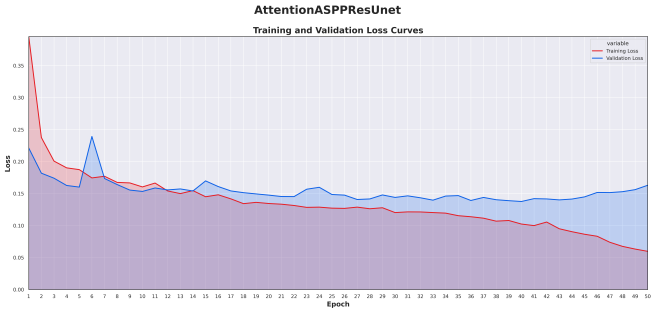


Fig. 3: Training and validation loss curves for Attention-Gated ASPP-ResU-Net. Training loss continues to decrease after roughly epoch 40 while validation loss plateaus, motivating early stopping with best-checkpoint selection.

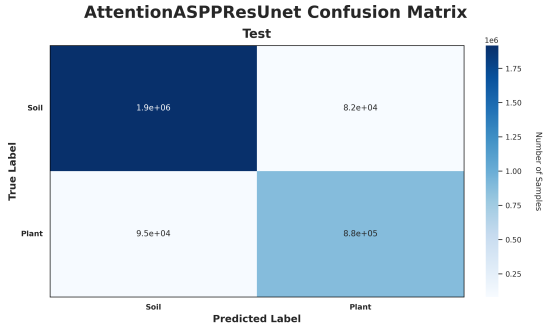


Fig. 4: Test-set confusion matrix for Attention-Gated ASPP-ResU-Net. False negatives on the plant class (95k) outnumber false positives on the soil class (82k), indicating that the residual errors are concentrated on plant-boundary pixels rather than on generic class confusion.

Adding ExG as an extra deep-learning input sometimes makes early convergence slightly faster, but it does not improve final segmentation quality. This is an informative negative result, because ExG is highly relevant to vegetation segmentation and might have been expected to help. Instead, the result suggests that the convolutional models are already able to learn similar colour-ratio cues directly from RGB input.

A learnable residual-scaling variant also underperforms the standard residual design. With low alpha initialization, early gradients become weaker and convergence slows down, leading to lower final performance than the standard residual block.

The learnable-residual variant reached a test Plant IoU of about 0.8188, clearly below the standard ResU-Net result (0.8318). These negative results matter because they clarify

TABLE IV: Negative-result ablations.

Ablation	Main observation	Outcome
ExG concatenated into deep input	Slightly faster early convergence, but lower final quality than RGB-only baseline	Not adopted
Learnable residual scaling (α)	Slower optimization and lower final performance than standard residual block	Not adopted

what the final gains are *not* coming from. In both cases, adding more structure or more prior information did not help. This reinforces the idea that the strongest improvements in this task come from effective feature propagation rather than simply introducing more components.

F. Robustness Analysis

The robustness experiments reveal a fundamentally different picture from the clean-set ranking, and contain the most practically significant finding of the study.

Under affine rotation, performance remains stable across all models, with only a modest monotonic decline as severity increases (AG-ASPP-ResU-Net Plant IoU 0.825→0.788). Under Gaussian blur, segmentation degrades moderately: for AG-ASPP-ResU-Net, Plant IoU falls from 0.832 to 0.784 at the highest severity, a decline that leaves the model still functional. JPEG compression shows a steeper trend, with Plant IoU dropping from 0.810 to 0.614, indicating that high-frequency boundary information matters more than it might appear. Brightness shift shows an asymmetric pattern: underexposure causes substantially more degradation than overexposure, and the resulting curve is non-monotonic, suggesting that the models are more sensitive to the loss of visible structure than to moderate saturation.

Gaussian noise, however, produces a qualitatively different and more severe failure. At level 5 (variance 0.2), Plant IoU for AG-ASPP-ResU-Net approaches near-zero—a near-complete collapse. More importantly, this vulnerability reveals a **ranking reversal** relative to clean-set performance. Between levels 2 and 3, ResU-Net Plant IoU collapses from approximately 0.56 to 0.11; by noise level 5, all residual-based models (ResU-Net, ASPP-ResU-Net, and AG-ASPP-ResU-Net) approach zero IoU. The plain U-Net, by contrast, retains a Plant IoU of approximately 0.27 at the same severity. On highly noisy inputs, the clean-set ranking is therefore fully inverted: the model with the worst clean-set performance

TABLE V: Summary of robustness behaviour under different corruptions (AG-ASPP-ResU-Net; noise comparison includes U-Net).

Corruption	Robustness	Key observation
Affine rotation	Strong	Modest monotonic decline (0.825→0.788)
Gaussian blur	Strong	Plant IoU 0.832→0.784 at highest severity
JPEG compression	Medium	Plant IoU 0.810→0.614; gradual boundary loss
Brightness shift	Uneven	Underexposure $\gg$ overexposure; non-monotonic
Gaussian noise	Weak	All residual models $\approx$0 at level 5; U-Net retains 0.27 (ranking reversed)

outperforms every model that surpassed it under ideal conditions. Figure 5 visualises this reversal: the residual backbone collapses between severity 2 and 3, whereas the plain U-Net curve degrades smoothly across the full range.

G. Failure Cases and Overall Findings

The failure cases bring the quantitative results into focus. The hardest examples usually involve low-contrast plant boundaries, dark illumination, or severe image degradation. These are not isolated mistakes, but recurring patterns that match the corruption experiments and confusion-matrix trends. Figure 6 illustrates a representative case in which thin leaf structures are partly missed and boundary segments drift from the ground-truth mask.

In total, the results reveal a consistent pattern. Classical methods remain useful as efficient references, but their dependence on handcrafted features limits their ceiling. Deep learning models are substantially stronger, and all deep models exceed the original paper benchmark. Most of the meaningful clean-set gain comes from better optimization and feature propagation rather than from simply increasing complexity. Attention-Gated ASPP-ResU-Net gives the best clean-set result, but the robustness experiments reveal a serious weakness: the residual-based designs that improve clean-set learning are also associated with substantially weaker robustness under severe Gaussian noise, producing a ranking reversal that has direct implications for real-world deployment.

V. Discussion

The results reveal a consistent pattern: the most reliable clean-set gains in this task come from improving feature propagation and optimization stability, rather than from increasing architectural complexity alone. This is most clearly reflected in the transition from U-Net to ResU-Net, which produces the largest single improvement among the deep models. By contrast, ASPP and attention gating still help, but mainly as refinements. Their contribution is real, yet much smaller than the gain introduced by residual learning. In this sense, the progression from U-Net to the strongest model is not simply a story of adding more modules, but of identifying which change addresses the main bottleneck most effectively.

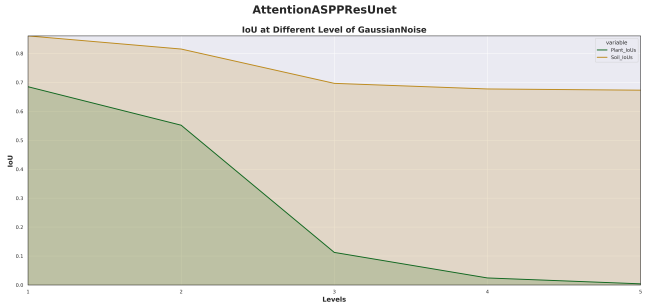
The comparison with classical methods helps clarify that bottleneck. ExG, Logistic Regression, and Random Forest remain meaningful baselines because they are fast, interpretable, and grounded in intuitively useful colour information. Their results show that vegetation can already be separated to some extent through handcrafted features, especially when HSV information is included. At the same time, their performance clearly plateaus below that of the encoder-decoder models. This suggests that the main advantage of the deep models is not just better colour discrimination, but the ability to use spatial structure, local texture, and boundary information in a more coherent way.

The negative results reinforce this interpretation. ExG concatenation into the deep models is a particularly informative failure case: although ExG is a highly relevant vegetation prior, adding it as an extra input channel does not improve final deep-model performance. This suggests that the convolutional backbone already learns similar colour-ratio cues directly from RGB data, consistent with the finding in the original paper [2]. A similar pattern appears in the learnable residual-scaling variant, which adds parameterization but does not outperform the standard residual design. Together, these failed modifications indicate that the gains in this project do not come from adding more information or more flexibility by default, but from changes that improve how useful features are learned and propagated.

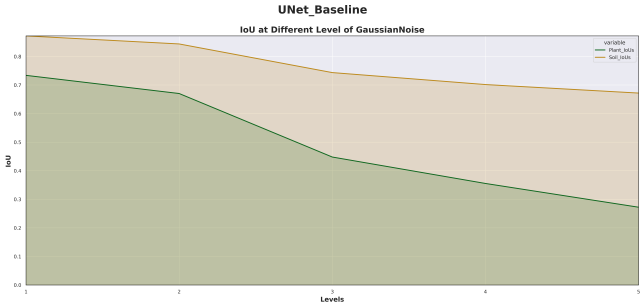
The robustness results add the most important qualification to the clean-set narrative. The ranking reversal under strong Gaussian noise is not a minor footnote: it reveals that the architectural change most responsible for clean-set improvement—the residual shortcut—is simultaneously associated with the greatest robustness failure. One plausible explanation is that the additive nature of residual connections allows noise injected at the input to propagate and accumulate across successive blocks, whereas the plain encoder of U-Net lacks this compounding path. GroupNorm, while effective for stabilizing training on clean data, may be misled by noise-corrupted activation statistics, potentially amplifying the problem at deeper layers. However, these are observations and candidate mechanisms, not confirmed causal claims: whether residual networks are categorically more or less noise-robust than plain networks is known to be task- and training-dependent, and targeted ablations would be required to isolate the contribution of each component. What the data do establish unambiguously is the ranking reversal itself, and its practical consequence: under severe Gaussian noise, architectural sophistication does not translate into robustness.

The validation-test gap is better understood as structural than incidental. The 1.2 percentage-point drop in Plant IoU from validation to test reflects a year-based distribution shift—training and validation images were collected in 2018–2020, while test images are from 2017—and is therefore an expected consequence of temporal domain shift rather than a sign of unstable training.

In summary, the experiments suggest that the key challenge in this task is not the absence of architectural complexity, but the difficulty of learning representations that are both accurate on clean data and stable under realistic corruption.



(a) Attention-Gated ASPP-ResU-Net: Plant IoU collapses after level 2.



(b) U-Net baseline: graceful, near-linear degradation, retaining Plant IoU ≈ 0.27 at level 5.

Fig. 5: Plant/Soil IoU under Gaussian noise across five severity levels. The residual-based architecture (left) and the plain U-Net (right) produce the ranking reversal discussed in the text.

(a) Ground truth. (b) Prediction.

Fig. 6: A representative failure case for Attention-Gated ASPP-ResU-Net on the test set. Thin leaf structures and low-contrast boundaries drive most of the remaining errors, consistent with the plant-side false-negative pattern in Fig. 4.

Further progress is unlikely to come simply from making the architecture more elaborate. More promising directions include training strategies that target noise sensitivity directly, such as noise-aware data augmentation, adversarial training, or architectural choices that decouple feature propagation from noise accumulation.

VI. Conclusion

The results show that deep encoder-decoder models are clearly more effective than classical pixel-wise baselines for plant-soil segmentation, with all deep models surpassing the original paper benchmark of Plant IoU 0.775. The clearest clean-set improvement comes from residual learning, while later additions such as ASPP and attention gating mainly act as smaller refinements.

The negative results are equally informative. Neither ExG concatenation nor learnable residual scaling improved the final deep-model performance, which shows that not every plausible enhancement is useful once the backbone is already strong. In that sense, the study does more than rank models: it helps clarify which kinds of modifications genuinely improve segmentation and which are largely redundant in this setting.

The main finding that goes beyond clean-set ranking is the robustness reversal under Gaussian noise. All residual-based models collapse near zero Plant IoU at high noise levels, while the plain U-Net retains 0.27 at the same severity. This reversal reveals a concrete accuracy-robustness trade-off associated with the residual-based architectures used in this study, and has direct implications for deployment in real agricultural environments where image quality is not guaranteed. The study is additionally limited to binary segmentation, which simplifies the agricultural scene and leaves richer forms of field understanding unexplored. Future work should prioritize training strategies that improve noise robustness without sacrificing the clean-set gains that residual learning provides, and consider extending the current pipeline beyond binary plant-soil separation toward more complete field understanding.

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